BankingSim

Detailed Project Report | Java Assignment - 2



Project	BankingSim – a Java banking simulator
Submitted by	Mohd Afnan
Roll number	160125733045
Course work	Java Assignment – 2
Platform	Java core with a browser-based web interface

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1. Introduction

BankingSim is a Java banking simulator in which customers open accounts, move money and review their transaction history, while an administrator manages the bank as a whole. The project began as a console program and has since grown into a web application that runs in the browser, while keeping the same Java core underneath. A central goal of the design is that account balances stay correct even when many operations happen simultaneously.

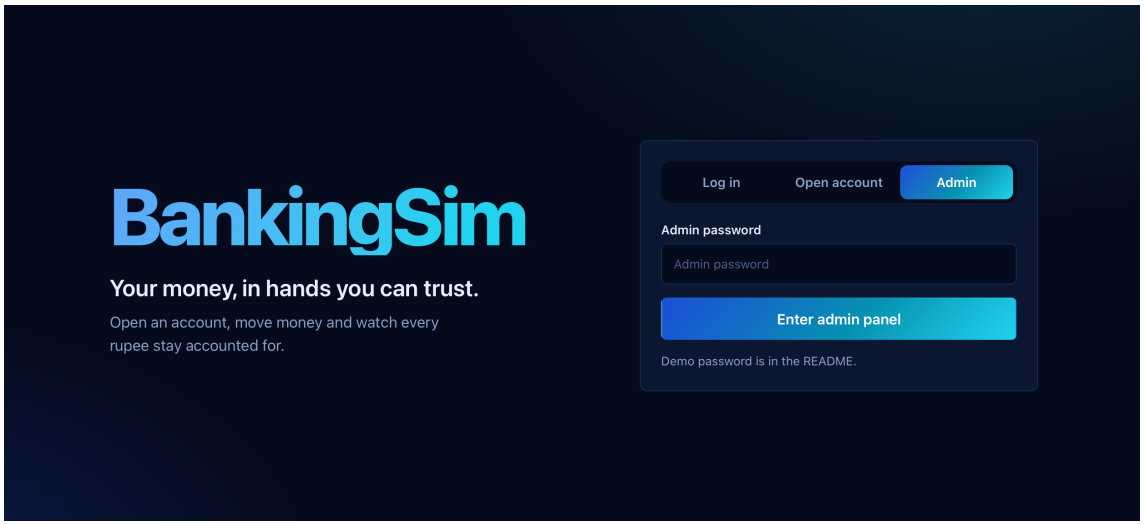


Figure 1: BankingSim landing screen showing the Admin sign-in tab (Log in, Open account and Admin tabs are available).

1.1 Objectives

- Model realistic banking behaviour with two account types that follow different rules.
- Give customers a simple, secure self-service flow: open an account, log in, deposit, withdraw, transfer, change PIN and review or download history.
- Give an administrator a separate panel to monitor, control and maintain the whole bank.
- Guarantee correct balances under concurrent use through thread-safe design.
- Handle mistakes gracefully with custom exceptions that produce clear, user-friendly messages.
- Persist all data between runs and offer timestamped backups.

1.2 Key features at a glance

Area	What the project provides
Account types	Savings (minimum balance Rs. 500, 4% yearly interest paid monthly) and Current (overdraft up to Rs. 10,000).
Customer features	Open an account, log in with a 4-digit PIN, deposit, withdraw, transfer to another account, change PIN, and view or download transaction history.
Security	Three wrong PINs lock the account; customers can reach only their own account; admin actions need a separate admin login.
Admin features	View totals and account mix, filter accounts by balance, lock or unlock accounts, pay interest, start an interest scheduler, create backups and run a thread-safety demo.
Thread safety	Balance-changing methods are synchronised; transfers lock accounts in a fixed order so two opposite transfers can never deadlock.

Area	What the project provides
Error handling	Custom exceptions for insufficient funds, invalid amounts, wrong PINs, locked accounts and unknown accounts.
Persistence	Accounts and transaction history are saved to a data file when the server stops and loaded at the next start; backups are available from the admin panel.

2. Account Types

BankingSim supports two kinds of account. Both share the same dashboard and operations, but each enforces its own balance rule when money leaves the account.

	Savings	Current
Opening deposit	Rs. 500 minimum	Rs. 1,000 minimum
Balance rule	Must stay at or above Rs. 500	Can go into overdraft, down to -Rs. 10,000
Interest	4% a year, paid monthly	None
Best for	Saving money	Everyday spending

2.1 Savings account

A Savings account earns 4% interest a year, credited one month at a time. It must always keep a balance of at least Rs. 500, so the amount a customer can actually take out is the balance minus Rs. 500. The dashboard shows this as the **Withdrawable now** figure. Each interest payment appears in the activity list as an *Interest* entry.

2.2 Current account

A Current account earns no interest and is designed for everyday spending. It allows an overdraft: the balance may fall below zero, down to a limit of -Rs. 10,000. The opening deposit is Rs. 1,000 minimum.

2.3 Account numbers

Every account receives a unique number when it is created, in the form **ACC1001**, **ACC1002** and so on. Customers use this number to log in and to receive transfers.

3. Customer Workflow

3.1 Opening an account

A customer opens an account by choosing Savings or Current, entering their name, setting a 4-digit PIN and making an opening deposit (Rs. 500 minimum for Savings, Rs. 1,000 for Current). The bank then creates the account with a unique number, records the opening deposit as the first transaction, and saves it. Afterwards the customer is taken to the login tab with the new account number already filled in.

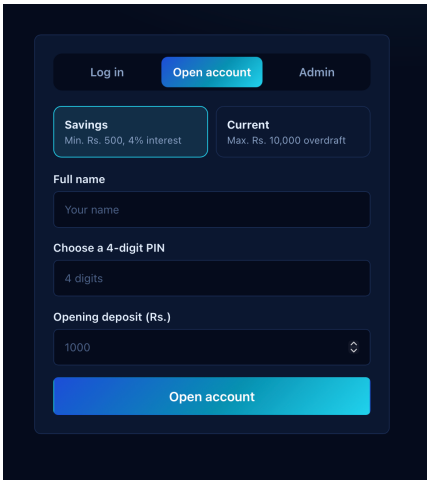


Figure 2: Open account form with account-type selection, full name, PIN and opening deposit.

3.2 Logging in

A customer logs in with their account number and 4-digit PIN. If the PIN is correct, the bank starts a session that lasts until the customer logs out or stays idle for 30 minutes, and they land on a dashboard that greets them by name. Each wrong PIN counts as a failed attempt and shows how many attempts remain. The third wrong PIN in a row locks the account until an admin unlocks it, and a correct PIN resets the count. Once logged in, a customer can only see and use their own account.

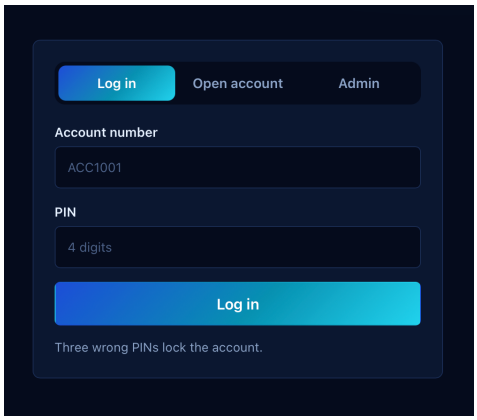


Figure 3: Log in tab. The hint below the button reminds users that three wrong PINs lock the account.

3.3 Dashboard

The dashboard has a left-hand menu (Overview, Deposit, Withdraw, Transfer, Transactions, Change PIN), a greeting by name, the available balance, the account type, the withdrawable amount, a bank-card style summary showing the account number, holder name and type, and a list of recent activity. In the sample session below, a Savings customer (Ben Pattrick, account ACC1001) has opened the account with Rs. 25,000, so the withdrawable figure is Rs. 24,500.

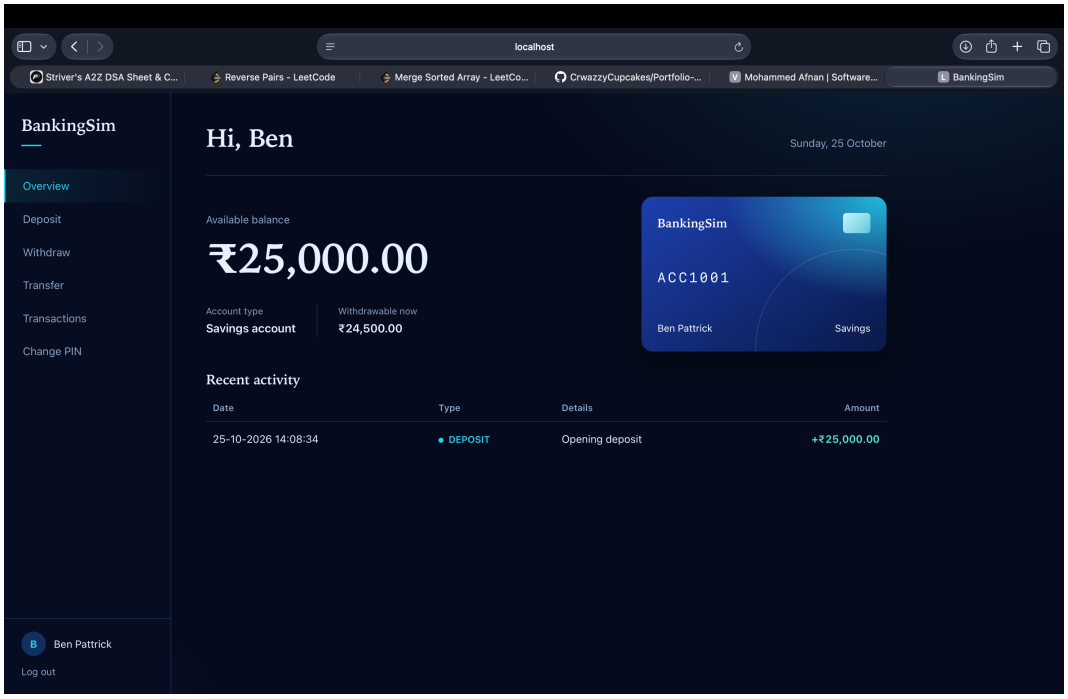


Figure 4: Customer dashboard for a Savings account.

3.4 Deposit

The customer enters an amount greater than zero, or taps one of the quick-fill buttons (500, 1,000 or 5,000). The money is added to the balance straight away, the new balance appears on the dashboard, and the deposit is recorded in the transaction history. Locked accounts cannot deposit.



Figure 5: Deposit screen with quick-fill buttons.

3.5 Withdrawal

The customer enters an amount, and the bank allows it only if the account's balance rule is met: a Savings account must keep at least Rs. 500, and a Current account can go into overdraft down to -Rs. 10,000. If the amount is too high, the customer sees a message telling them the most they can withdraw. Locked accounts cannot withdraw.



Figure 6: Withdraw screen. The text reminds the user to stay within the withdrawable amount.

3.6 Transfer

A customer can transfer money to another account by entering its account number (for example ACC1002) and an amount. The transfer is refused if the amount is too high, the destination account does not exist, or either account is locked.

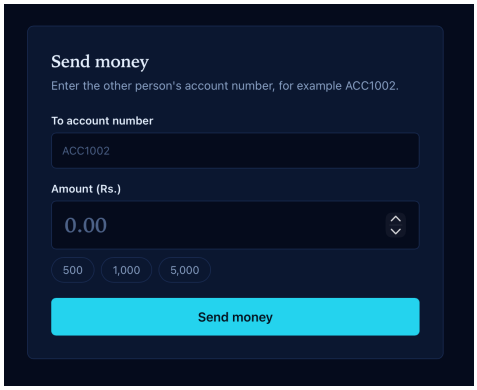


Figure 7: Send money screen.

3.7 Change PIN

A customer can change their PIN by entering the current PIN and a new 4-digit one. The screen advises choosing a PIN that is not used anywhere else.

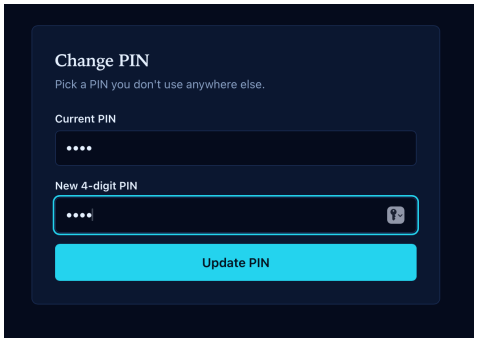


Figure 8: Change PIN form.

3.8 Transaction history

The transaction history lists every operation with its date, type, details, amount and the balance afterwards. It can be sorted (for example newest first) and downloaded as a CSV statement. The sample below shows how a running balance is maintained across deposits and withdrawals.



Figure 9: All transactions view with sorting and Download CSV.

#	Type	Details	Amount (Rs.)	Balance (Rs.)
1	Deposit	Opening deposit	+25,000.00	25,000.00
2	Deposit	Cash deposit	+50,000.00	75,000.00
3	Withdrawal	Cash withdrawal	-25,000.00	50,000.00
4	Deposit	Cash deposit	+1,20,000.00	1,70,000.00
5	Deposit	Cash deposit	+5,500.00	1,75,500.00
6	Withdrawal	Cash withdrawal	-1,00,000.00	75,500.00

Table: Sample ledger from the screenshot (amounts in Indian digit grouping).

4. Security

Security in BankingSim is built from a few simple, strictly enforced rules:

- **PIN authentication:** each account is protected by a 4-digit PIN that is required at login and when changing the PIN.
- **Account lock-out:** three wrong PINs in a row lock the account. A locked account cannot log in, deposit, withdraw or take part in transfers until an administrator unlocks it. A correct PIN resets the failed-attempt count.
- **Session handling:** a session starts after a successful login and ends on log out or after 30 minutes of inactivity.
- **Account isolation:** a logged-in customer can only reach their own account.
- **Separate admin login:** administrator actions require a different sign-in with a single admin password, so customer credentials never grant admin access.

5. Administrator Panel

The administrator signs in from the Admin tab with a single password and gets a separate panel for managing the whole bank.

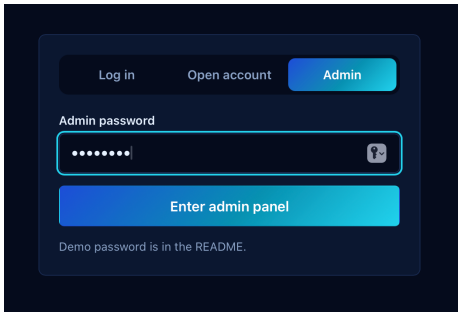


Figure 10: Admin sign-in.

5.1 Admin capabilities

- **Overview:** total accounts, total holdings, the Savings and Current split, and the highest balance.
- **Accounts list:** view all accounts and filter them by balance.
- **Lock / unlock:** lock or unlock any account, including accounts locked by failed PIN attempts.
- **Interest:** pay one month of interest to all unlocked Savings accounts immediately.
- **Interest scheduler:** start a background scheduler that pays interest automatically. It runs as a background Java thread where 10 seconds represents 1 month.
- **Backups:** create a timestamped backup of all bank data.
- **Thread-safety demo:** run real Java threads against throw-away accounts to demonstrate correct behaviour under concurrency without touching any customer data.

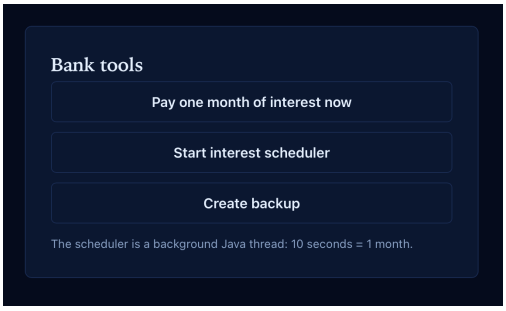


Figure 11: Bank tools: pay interest, start scheduler and create backup.

6. Thread Safety and Concurrency

A banking system must keep balances correct when several operations arrive at the same time. BankingSim addresses this at the level of the Java core:

- **Synchronised methods:** every balance-changing method (such as deposit and withdraw) is synchronised, so two threads can never update the same balance at once and cause a lost update.
- **Ordered locking for transfers:** a transfer touches two accounts, so it locks them in a fixed order. Because every transfer acquires locks in the same order, two opposite transfers (A to B and B to A at the same moment) can never deadlock.
- **Background scheduler:** the interest scheduler is a genuine background Java thread, which runs alongside normal customer activity.
- **Demonstration:** the built-in demo uses real threads on temporary accounts so the behaviour can be shown safely at any time.

> **Why fixed-order locking works:** deadlock needs a cycle in which each thread holds one lock and waits for the other. When all threads take locks in the same order, such a cycle cannot form.

7. Error Handling

Instead of failing silently or crashing, BankingSim defines custom exceptions for each kind of mistake and converts them into clear messages for the user.

Situation	Exception category	What the user experiences
Withdrawal or transfer exceeds what the account rule allows	Insufficient funds	A message telling the customer the most they can withdraw.
Zero or negative amount entered	Invalid amount	The operation is rejected with an explanation.
Incorrect PIN	Wrong PIN	Shows how many attempts remain.
Account is locked	Locked account	Deposit, withdrawal, login and transfers are refused.
Destination account number does not exist	Unknown account	Transfer is refused.

8. Data Persistence and Backups

All accounts and their full transaction history are saved to a data file when the server stops and are loaded again at the next start, so nothing is lost between runs. In addition, the administrator can create timestamped backups of all data from the admin panel at any time. Customers can also export their own history as a CSV statement.

9. Summary of Business Rules

Rule	Value
Savings opening deposit	Rs. 500 minimum
Current opening deposit	Rs. 1,000 minimum
Savings minimum balance	Rs. 500 (withdrawable = balance - Rs. 500)
Current overdraft limit	Down to -Rs. 10,000
Savings interest	4% a year, paid monthly
PIN format	4 digits
Failed PIN limit	3 in a row locks the account (admin must unlock)
Session timeout	30 minutes idle
Scheduler timing	10 seconds = 1 month
Quick-fill amounts	500, 1,000, 5,000
Account number format	ACC1001, ACC1002, ...

10. Conclusion

BankingSim shows that a small, careful Java core can power a real-feeling banking app. Savings and Current accounts follow their own rules, balances stay correct even when many operations happen at once, and custom exceptions turn mistakes into clear messages. The same core that ran in the console now serves a clean web interface for customers and an administrator. All accounts and transaction history are saved to a data file when the server stops and loaded again at the next start, with timestamped backups available from the admin panel, so nothing is lost between runs.